SiteProof

Construction & Mining Editions

Implementation, Domain & Feature Reference — Revision 4

Prepared for: Lisan Al-Gaib, Self Hawk Tech (Pty) Ltd **Date:** 22 September 2026 **Supersedes:** All earlier SiteProof documents (Revisions 1–3)

Scope of this revision

SiteProof is now defined as **one safety platform with two industry editions: Construction and Mining.** Nothing else is in scope.

Petrochemical has been dropped. South Africa now has only three operating fuel production facilities — Astron Energy in Cape Town, and Sasol's Natref and Secunda plants (TopAuto, Engineering News). That is too few principals, the contractor work is episodic, and its signature workflow — permit-to-work — carries a safety-critical liability out of proportion to the market.

The focus is safety. Both editions lead with daily safety capture. Construction keeps its delay-claims module as a secondary, commercial layer, because contractual notice regimes are a construction-specific problem. Mining does not get an equivalent — mining contractors' commercial disputes are handled by in-house commercial teams under bespoke contracts, and chasing them would dilute the safety focus.

What carries forward from Revision 3: the safety-first sequencing, the FEM correction, the injury-on-duty claim lifecycle, the POPIA position including the 2026 Health Information Regulations, and Stitch as the payment provider. These are restated where they apply, so this document stands on its own.

How to read this document

Part	Content
1	The platform model: editions on a shared core
2	The shared core, in detail
3	The Construction edition
4	The Mining edition
5	Implementation: stack, data model, phases
6	Complete feature inventory — core, Construction, Mining
7	Resource library — by edition
8	Risks and caveats

Phase markers. Core and Construction features use **P1–P4** as before. Mining features use **M1–M3**, because the mining edition starts only once Construction is paying. See section 5.5.

On verification. Every legal and regulatory statement was checked against a source at the time of writing, and sources are linked. Earlier revisions each contained something later corrected, and the mining material in this revision is newer and less tested than the construction material. Verify at source before writing code that depends on anything here.

This is not legal advice. Mining in particular is a heavily regulated, safety-critical environment. Before building the Mining edition, have its design reviewed by a mining SHE practitioner, a mine's contractor management function, and a privacy practitioner.

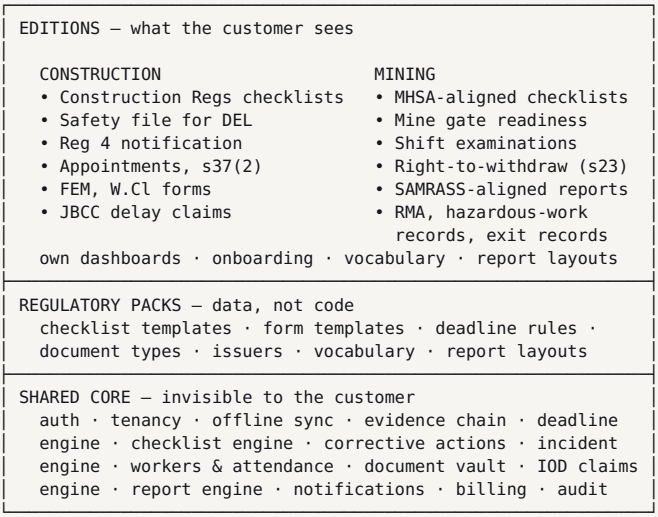
PART 1 — THE PLATFORM MODEL

1.1 One platform, two editions

A customer signs up, chooses **Construction** or **Mining**, and lands in a dashboard built specifically for that industry. A mining contractor never sees a JBCC field or a Regulation 4 notification. A construction subcontractor never sees a shift examination or a SAMRASS classification.

Underneath, both editions run on one shared core. That is the leverage: each edition feels like a specialist product, while you maintain one codebase.

1.2 The three layers



The rule that keeps this healthy: editions depend on the core; **the core never knows editions exist.** The moment core code contains `if edition == 'mining'`, the model starts to rot. Edition behaviour lives in edition modules or in regulatory-pack data.

1.3 The edition attaches to the site, not the organisation

The signup choice sets the **primary** edition and lands the user in that dashboard. But the edition is recorded on each **site**, not locked to the organisation, because the same contractor frequently works both industries — an electrical, scaffolding or steel-erection contractor may be on a building site in March and a mine's surface plant in June.

- Each site belongs to exactly one edition and uses that edition's rules, forms and workflows.
- An organisation can add the second edition later without a new account.
- **Company-wide records are shared across editions:** the worker register, competencies, medical fitness status, statutory documents and the injury-on-duty claims. A worker's certificates follow them from the building site to the mine.

That last point is a real advantage. A single company-wide compliance record spanning both industries is something no single-industry competitor offers.

1.4 Regulatory regimes as data

Each edition is backed by one or more **regulatory regimes**. Everything regime-specific is data keyed to the regime.

```
CREATE TABLE regulatory_regimes (
    id            uuid PRIMARY KEY,
    code          text UNIQUE NOT NULL,  -- 'za_construction_2014', 'za_mhsa_1996'
    name          text NOT NULL,
    edition       text NOT NULL,         -- 'construction' | 'mining'
    jurisdiction  char(2) NOT NULL DEFAULT 'ZA',
    version       text NOT NULL,
    effective_from date NOT NULL
);

-- regime-keyed content
regime_checklist_templates  (regime_id, template_id)
regime_document_types       (regime_id, doc_type, default_issuer, validity_days)
regime_deadline_rules       (regime_id, rule_code, period, unit, basis, source_ref)
regime_vocabulary           (regime_id, term_key, label)
regime_report_layouts       (regime_id, report_type, template_id, version)
regime_incident_taxonomy    (regime_id, code, label, reportable, report_rule_code)

-- binding
sites.regime_id  uuid NOT NULL REFERENCES regulatory_regimes(id)
organisations.primary_edition text NOT NULL
organisations.enabled_editions text[] NOT NULL
```

Two consequences. A regulatory change — a new threshold, a new mandatory code of practice guideline, a renumbered form — becomes a data update rather than a code release. And adding the Mining edition becomes largely a content project plus a handful of genuinely new modules, not a rebuild.

1.5 Positioning per edition

Construction: *keeps South African contractors compliant, gets their injury claims paid, and stops them losing delay claims — built for the Construction Regulations, COIDA and JBCC.*

Mining: *gets your people through the mine gate ready to work, keeps your contractor safety records the way the mine and the inspectorate expect them, and holds one compliance record across every mine you work on.*

Never say "safety checklist app", "all-in-one", or "digital transformation". Specificity is the moat; generality is how SafetyCulture beats you.

1.6 Who buys each edition

	Construction	Mining
End user	Subcontractor, CIDB grade 2–5	Mining contractor: drilling, development, shaft work, engineering maintenance, electrical, steel, transport, surface plant services
Principal	Principal contractor	The mine
Channel	SACPCMP-registered CHSOs; Master Builders	Contractor SHE managers; mine contractor-management coordinators
Primary pain	Safety file on demand; LGS lapses; stalled injury claims; lost delay claims	Getting workers cleared at each mine; mine-specific medicals and inductions; producing records the mine and inspectorate require
Willingness to pay	Low–moderate	Moderate–high

1.7 Pricing

Edition	Tier	Indicative	Core inclusion
Construction	Site	R249–R399/mo	Safety, incidents, diary, safety file
Construction	Contractor	R699–R999/mo	+ IOD lifecycle, delay claims, statutory vault
Construction	Principal Contractor	R2,500–R6,000/mo per active site	+ sub accounts, compliance rollup
Mining	Contractor	R1,500–R3,500/mo	Safety, incidents, workers, mine readiness, RMA IOD
Mining	Contractor Plus	Priced per active mine	+ multi-mine readiness, hazardous-work records, exit records
Both	Add-on edition	Discounted second edition	Shared company records across both

Mining is priced higher because the per-worker compliance burden is heavier and the contractors are typically larger. **These are placeholders to be tested in validation, not decided prices.**

1.8 Payments

Stitch as primary, with DebiCheck as the default method and card as the fallback. Stitch supports every mandated recurring payment method in South Africa, including DebiCheck, with automated fallback retry via an alternative method (Stitch). DebiCheck mandates are created as Payment Consent Requests via a GraphQL API and, once granted, collected against (Stitch docs); the collection process itself runs over an SFTP-based file exchange (Stitch docs). Budget integration time accordingly.

Build a **provider-agnostic billing layer** — plans, subscriptions, mandates, invoices and payment attempts — behind a `PaymentProvider` interface, so provider decisions never touch business logic. Billing is per organisation, with editions and sites as line items. **Read Stitch's current API and webhook references directly before implementing; do not build from this summary.**

PART 2 — THE SHARED CORE

Everything in this part is edition-agnostic. Each item is configured by regulatory-pack data or extended by edition modules.

2.1 Evidence engine

The single most important core capability, because every safety record, incident and claim rests on it.

- **In-app capture** with GPS, device timestamp, site, user and device identifier
- **On-device compression** to a 2048px long edge before hashing
- **SHA-256 at capture**, re-verified server-side on receipt
- **Dual timestamps** — device and server receipt — with any divergence shown honestly, never hidden
- **Per-project hash chain:** `SHA256(content_hash || server_received_at || site_id || prev_chain_hash)`, making insertion or deletion detectable
- **Daily signed chain anchor** written to a separate append-only store
- **Device attestation** via Play Integrity and App Attest
- **Mock-location flagging**, recorded rather than discarded
- **Immutable originals** — markup and watermarks are derived files
- **Append-only access log** on every view, export and share

Build to the factors in section 15 of the Electronic Communications and Transactions Act 25 of 2002 — reliability of generation, storage and integrity maintenance. **Read section 15 at source. Never market the system as "tamper-proof" or "court-admissible."**

2.2 Offline-first sync

- Client-generated UUIDs, making sync idempotent
- Separate metadata and media queues, so records reach the server before their photos
- Resumable chunked uploads via S3 multipart
- **WiFi-preferred by default**, with an explicit "upload now on data" button and size estimate
- Local storage budget with loud warnings before the queue fills
- Per-record sync status
- Append-only where possible; hard lock on anything linked to a submitted claim, closed safety file or statutory report

Mining raises the bar. An underground shift can run eight to twelve hours with no connectivity at all, and sync may only happen on surface. The core must treat a full shift offline as normal. See section 4.10 for the device constraints that follow.

2.3 Deadline engine

One component serving every statutory and contractual clock in both editions.

Must handle: calendar days versus working days, per rule; South African public holidays (1 January, 21 March, 27 April, 1 May, 16 June, 9 August, 24 September, 16 December, 25 December, 26 December, plus Good Friday and Family Day); the rule moving a Sunday holiday to the Monday; ad hoc proclaimed holidays; project-specific shutdowns such as the construction builders' holiday; and recurring cadences such as monthly medical reports.

The calendar is a table, never a constant. Rules come from `regime_deadline_rules`.

Liability protections: show the rule and its source; let users inspect excluded days; never present a deadline as authoritative; log every computation and notification; and alert on **job non-completion**, not only failure — a silently stopped deadline job is a business-ending defect.

2.4 Checklist engine

- Versioned templates supplied by regulatory packs, with organisation-specific additions
- **Photo mandatory on fail, optional on pass**

- **Reference photos** per item showing what "correct" looks like
- Response types: pass/fail/N-A, scale, numeric reading, count, text
- Scheduling: per shift, daily, weekly, per activity, ad hoc
- Every fail can raise a corrective action
- Template version stored on each completed instance, so a 2027 checklist is reproducible exactly in 2030

2.5 Corrective action register

Owner, due date, severity, source record, before and after photographs, close-out verification. **Open-items ageing** is the core safety metric for both editions and the one a principal — contractor or mine — will judge performance on.

2.6 Incident engine

Captures once and routes to every obligation the site's regime triggers. The **incident taxonomy** (types, reportability, reporting rules) comes from `regime_incident_taxonomy`, which is how the same engine produces OHSA section 24 reporting on a construction site and SAMRASS-aligned reporting on a mine.

Near misses get a one-tap, thirty-second, no-blame entry in both editions.

2.7 Workers and attendance

- Worker register with encrypted ID number (last four digits shown by default)
- Credentials and competencies with expiry
- Medical fitness **status and expiry only** — see 2.10
- **Site-specific clearance** — a worker can be cleared for one site and not another. Minor in construction, central in mining
- QR/NFC worker cards scanned by the foreman or supervisor; roll-call fallback
- Geofence check, server timestamp, and a headcount reconciliation flag
- **Append-only attendance events**; corrections as new rows
- Clock-in block when a required credential, medical or induction has lapsed
- Export for payroll; never become payroll

2.8 Statutory document vault

Organisation-level documents with issuer, reference, issue and expiry dates, verification status and escalating expiry alerts at 60, 30 and 7 days. **Issuer-aware:** FEM, Compensation Fund or RMA for Letters of Good Standing.

2.9 Injury-on-duty claims engine

The Revision 3 lifecycle, generalised: claim stage tracker, document checklist, account-free doctor upload links, monthly progress-report cadence, claim-number capture, attendance-triggered resumption report, light duties, reimbursement ledger and submission-pack export.

Insurer routing comes from the organisation profile: FEM for most construction employers, RMA for most mining employers, the Compensation Fund otherwise. Form templates and submission instructions are regime and insurer data. See 3.8 and 4.8.

2.10 Data protection

Rule 1 — no biometrics. Ever. Section 26 of POPIA prohibits processing special personal information, including health and biometric information, unless an exception applies (POPIA s26). QR and NFC do the attendance job.

Rule 2 — medical fitness: status, not contents. Store the certificate's existence, dates, practitioner reference and fit / fit-with-conditions / unfit. Never findings, diagnoses or restrictions.

Rule 3 — IOD medical reports are health data held under section 32(1)(f). That section authorises employers to process health information where necessary to implement laws creating rights dependent on health, or to support workers entitled to benefits for sickness or work incapacity, subject to an obligation of confidentiality under section

32(2) (POPIA s32). The **Health Information Regulations, 2026** came into force on publication on 27 February 2026 with no grace period and cover employers and their operators (ITLawCo, Bowmans). SiteProof is an operator. Controls: separate encryption key, claims-handler role only, never visible to principals, purpose-limited, fully logged.

Rule 4 — mining raises the stakes. Mining work generates far more health-adjacent data: records of hazardous work, exposure measurements linked to individuals, exit records. Section 4.9 sets out what the Mining edition may and may not hold.

2.11 Report engine

Versioned templates, customer branding, asynchronous generation for large packs, and a manifest with hashes on every export. The construction **safety file** and the mining **readiness and compliance pack** are both products of this engine.

2.12 Roles

Role	Capabilities
Supervisor / foreman	Checklists, examinations, hazards, near misses, incidents, attendance, photos. No health documents after capture.
SHE officer / CHSO	All safety records, appointments, worker register, packs. Health documents only with the claims-handler capability.
Claims handler	IOD claims including health-flagged documents
Contracts manager / QS	Construction delay claims (Construction edition only)
Organisation owner	Everything, billing, users
Bookkeeper	Billing, reimbursement totals — never medical documents
Principal (PC or mine)	Scoped read of contractor compliance status. **Never** draft claims, **never** health information
Treating doctor	Account-free, upload-only, single claim

2.13 Security

TLS with HSTS; separate encryption keys for evidence, application data and health data; S3 Object Lock on evidence; no public object URLs; append-only audit log shipped off-instance; signed, idempotent payment webhooks; secrets in a managed store; deadline-job non-completion alerting; tested quarterly restore; tenant isolation in the data layer plus PostgreSQL row-level security.

PART 3 — THE CONSTRUCTION EDITION

Restated from Revision 3 and condensed. The construction material has been through three revisions; the detail stands.

3.1 Statutory stack

- **Occupational Health and Safety Act 85 of 1993** — section 24 reportable incidents; section 37(2) mandatary agreements
- **Construction Regulations 2014** — the documentation spine (full text (PDF), ASAQS guidelines (PDF))
- **COIDA 130 of 1993** — injury claims; most construction employers insured with **FEM**
- **POPIA** and the 2026 Health Information Regulations

3.2 Regulations 3 and 4

Regulation 3 — construction work permit above the threshold, raised by exemption notice to above forty million rand for work from 7 February 2017 (Labour Guide). **Regulation 4** — for all other work, notify the provincial director **at least 7 days before** work begins (Regulation 4). Store thresholds as data.

3.3 Safety modules

Construction Regulations checklist library · corrective actions with ageing · **housekeeping walks** scored by zone and trended · toolbox talks with content library · hazard identification · PPE checks and issue register · risk assessments with review dates · **appointments register** with letters and gap detection · **Section 37(2) mandatary agreements** · induction records · **safety file assembly** in inspector order · inspection readiness view.

3.4 Medical certificates of fitness

Regulation 7(8) requires every employee to hold a valid medical certificate of fitness specific to the work, issued by an occupational health practitioner on **Annexure 3** (Annexure 3 (PDF)), valid for one year (example (PDF)). Features: expiry tracking with clock-in block; pre-filling the employer's section from trade and risk assessment. Status only, never contents.

3.5 Incidents — three paths

COIDA — report within 7 days on W.Cl.2; failure is an offence (Bowmans). **OHSA section 24** — death, unconsciousness or disablement reportable to the Department of Employment and Labour on Annexure 1 (MAKROSAFE). **Internal** — investigation and incident register (Gawie le Roux Institute).

3.6 Letter of Good Standing

Usually from **FEM** for construction (FEM, Tenderkom). A gating document for site access and tenders.

3.7 Workers and attendance

Foreman-driven QR/NFC scanning; append-only events; headcount reconciliation; MCoF clock-in block.

3.8 Injury-on-duty lifecycle

Form	Purpose	When
W.Cl.2	Employer's report (Part A); Part B to the doctor	Within 7 days
W.Cl.3	Employee's report	With the claim
W.Cl.4	First medical report	First consultation

W.Cl.55 / 56	Acknowledgement / acceptance, with claim number	From the insurer
W.Cl.5	Progress report monthly; final report on the same form	During absence; at recovery
W.Cl.6	Resumption report, with the final W.Cl.5	On return to work

Only after every form is submitted is the case closed and paid (Labour Guide, Labour Guide claims guide). The employer carries compensation for the first three months and reclaims it (Bowmans) — so open claims are shown as **money owed**. No public submission API has been found for CompEasy or FEM; the product prepares, the employer uploads.

3.9 Site diary

Two-tier entry: sixty-second default, escalation when something is claimable. Weather recorded by **effect** (activities stopped, hours lost, critical path, delayed effect), labour by trade and subcontractor, plant working versus standing, materials expected but not received.

3.10 Delay claims — the secondary layer

Kept in the Construction edition as a commercial layer on top of the safety record.

JBCC notice regime: notice of intention within **20 working days** of awareness, or the claim is unenforceable (Cox Yeats, Hahn & Hahn); quantification within **40 working days**; principal agent response within **20 working days**, with silence as deemed rejection (Agora Africa, JBCC advisory note (PDF)). Clause numbers are Edition 6.2; verify the current edition.

Features: claims register, three-clock state machine, notices emailed from the contractor's own domain with proof of service, confirmation of verbal instruction, evidence linking, quantum entry, submission and dispute packs, programme chart rendering.

3.11 Vocabulary

Site diary (not daily log) · contract instruction (not change order) · principal agent · principal contractor · employer (JBCC) / client (Construction Regulations) · practical completion · snag list · preliminaries and generals · bill of quantities.

3.12 Validation

Three CHSOs, five subcontractors, three principal contractors. Exit signal: **a CHSO would put their client base on it.**

PART 4 — THE MINING EDITION

This is the new material in Revision 4 and the least tested. Read it as a well-sourced starting hypothesis, then validate it with the people in section 4.14 before building.

4.1 How mining differs from construction

	Construction	Mining
Governing Act	OHSA 85 of 1993 + Construction Regulations	**Mine Health and Safety Act 29 of 1996** + Mine Health and Safety Regulations
Regulator	Department of Employment and Labour	**Mine Health and Safety Inspectorate**, Department of Mineral and Petroleum Resources
Unit of work	Working day	**Shift**
Location model	Site	Mine → shaft or pit → level or section → working place; surface vs underground
Principal	Principal contractor	**The mine**
Worker clearance	Largely per company	**Per mine** — medical, induction and competencies to that mine's standard
Injury insurer	Mostly FEM	Mostly **Rand Mutual Assurance (RMA)**
Occupational lung disease	COIDA	**Occupational Diseases in Mines and Works Act (ODMWA)** — separate system
Worker voice	Limited	**Statutory health and safety representatives and committees**; right to leave dangerous working places
Connectivity	Usually some	Underground: often none for a full shift

4.2 The regulatory spine: the Mine Health and Safety Act

The MHSA regulates employers' and employees' duties to identify hazards and eliminate, control and minimise risk, and entrenches the right to refuse to work in dangerous conditions (gov.za). The consolidated Act is on SAFLII; the Mine Health and Safety Council publishes the Act and Regulations as a single booklet (PDF).

The sections that drive product features:

Section	Obligation	What SiteProof does
s9 Codes of practice	Employers must prepare and implement mandatory codes of practice that comply with guidelines issued by the Chief Inspector of Mines; failure is a contravention and a criminal offence	COP register and worker acknowledgement against the mine's COPs (4.5)
s10 Training	Health and safety training	Training and induction records per worker, per mine
s11 Risk assessment	Identify hazards, assess risk, and implement control measures **in the order listed** in the Act; periodically review including hygiene and medical surveillance results	Risk assessment tool that enforces hierarchy-of-control ordering (4.7)
s11(5) Investigations	Commence an investigation **within 10 days** of an accident, serious illness or health-threatening occurrence, with the health and safety representative; deliver the report **within 30 days** to the Principal Inspector of Mines and the health and safety committee	Two deadline clocks on every qualifying incident (4.6)
s11(8) Scene preservation	After a fatal or likely-fatal incident or loss of a limb, the site may not be disturbed without the Principal Inspector's consent	Scene-preservation flag and evidence capture prompt
s12 Occupational hygiene	Measurements recorded so they can be linked to each employee's medical surveillance record	Exposure category and hazardous-work history per worker (4.9)
s13 Medical surveillance	Initial and periodic medical examinations for employees exposed to health hazards	Fitness **status and expiry** per worker, per

		mine
s14 Record of hazardous work	A service record, in the prescribed form, of employees doing work subject to medical surveillance	Hazardous-work service record (4.9)
s15–17	Record of medical surveillance, annual medical reports, **exit certificates**	Tracks that exit examinations happened — not their contents
s23 Right to leave	Employees may leave a working place posing serious danger; employers must set procedures after consulting the health and safety committee	Right-to-withdraw register (4.4)
s25 H&S representatives	Representatives and committees; employer must notify them of accidents and inspections	Representative linkage and notifications (4.8)

Sources for the section content: MHSA text (CER, PDF), SAFLII, section 23 annotated, section 25 annotated. The Act also requires the records under sections 12(3), 13(3)(c) and 14(1) to be **retained until the mine closes** (MHSA text) — which makes your retention design a statutory question, not a preference.

4.3 The Regulations

The Mine Health and Safety Regulations run to 23 chapters, including Appointments and Administration, Explosives, Machinery and Equipment, Occupational Hygiene, Occupational Health, **Qualifications and Competencies**, **Rescue, First Aid and Emergency Preparedness**, Forms, and **Reporting of Accidents and Dangerous Occurrences** (SAFLII).

Chapter 23 — reporting is the one that most shapes the incident engine. Dangerous occurrences must be reported **immediately by the quickest means available** and confirmed without delay on **Form SAMRASS 1**; heat-stroke or heat-exhaustion injuries also require **SAMRASS 8** within 14 days (Regulation 23). SAMRASS — the South African Mines Reportable Accidents Statistics System — has a published **codebook** classifying accidents, causes and locations (SAMRASS codebook (PDF)).

Design consequence: load the SAMRASS classification into `regime_incident_taxonomy` so that incident capture on a mine site produces SAMRASS-aligned data from the start. The mine will almost always do the formal report — but a contractor who hands the mine a correctly classified, evidenced record within the hour is a contractor the mine wants back.

4.4 Signature workflow 1 — Right to withdraw (section 23)

The feature that most clearly marks this as a mining product. Nothing like it exists in the Construction edition, because nothing like it exists in construction law in the same form.

An employee may leave a working place where circumstances, with reasonable justification, appear to pose a serious danger; a health and safety representative may direct them to. The employer's procedures must provide for notifying supervisors and representatives, joint participation in resolving the issue, involving an inspector where necessary, and where appropriate, assigning the employee to suitable alternative work (section 23 annotated, MHSA text).

What the product records:

```
withdrawals        id, site_id, working_place_id, occurred_at,
                   initiated_by,          -- employee | hs_rep
                   worker_ids[], hs_rep_id, danger_description,
                   photo_ids[], supervisor_notified_at,
                   hs_rep_notified_at, inspector_involved,
                   alternative_work_assigned, resolution,
                   resolved_at, returned_at, status
```

Why a contractor wants this: a withdrawal stops work, and stopped work is disputed time. A timestamped, evidenced record of who withdrew, why, who was notified and how it was resolved protects the contractor commercially and demonstrates to the mine that the statutory procedure was followed. It also protects workers — the record exists whether or not anyone later disputes it.

Design principle: the app must make withdrawing *easy and blame-free* from the worker's side. A feature that makes withdrawal feel like paperwork punishment would be both unethical and a statutory problem.

4.5 Signature workflow 2 — Mine gate readiness

The mining edition's core commercial problem. Practitioner guidance states that the MHSA applies to every person working on mine premises, that contractors on a mine are treated as the mine's employees for health and safety purposes, and that medicals and induction are therefore required — with many mines operating a short-duration permit of around five to seven days for brief work, subject to a short visitor induction (MiningSafety). **That is a practitioner explanation, not the statute — verify how each target mine applies it.**

Each mine sets its own standards. The DMR guideline on **minimum standards of fitness to perform work at a mine** has the occupational medical practitioner determine minimum standards per job category (DMR guideline (PDF)). So a worker fit for one mine is not automatically fit for another.

The readiness matrix:

	Mine A	Mine B	Mine C
Worker 1 (driller)	✔ ready	✖ induction missing	◐ medical exp. 14 days
Worker 2 (fitter)	✔ ready	✔ ready	✖ competency

Per worker, per mine: mine-specific induction (date, version, expiry); medical fitness to that mine's standard (status and expiry only); competencies required for the role at that mine; appointments where the role requires one; PPE issued; any short-duration permit and its expiry.

Per contractor, per mine: the mine's contractor documentation requirements; RMA Letter of Good Standing; the contractor's own safety documentation the mine requires.

Outputs: a readiness status the contractor sees before mobilising anyone; a pre-arrival pack exported to the mine's contractor-management office; expiry alerts **per mine**, because a worker can lapse at one mine while remaining valid at another.

Why this beats mine-side systems: mine-side onboarding platforms exist and sell to mines (OPMS, Kynection, Quartex, Ariscu). Each shows the mine its own gate. **None gives the contractor one view across every mine it serves.** That is SiteProof's position: contractor-side, multi-mine, exporting to whatever each mine demands.

4.6 Signature workflow 3 — Incidents on a mine

Built on the core incident engine with the mining regime's taxonomy and clocks:

Trigger	Clock	Source
Dangerous occurrence	**Immediately** by quickest means, then SAMRASS 1	Reg 23.5
Heat-related injury	SAMRASS 8 within **14 days**	Reg 23
Accident, serious illness or health-threatening occurrence	Investigation starts within **10 days**	s11(5)(aA)
Same	Investigation report within **30 days** to Principal Inspector and H&S committee	s11(5)
Fatal, likely fatal, or loss of limb	**Scene preserved** pending Principal Inspector consent	s11(8)
Any accident	Notify H&S representatives as soon as practicable	s25

The "immediately" obligation needs a different UI from the construction 7-day clock. It isn't a countdown; it's a prompt: *call now*, with a one-tap log of who was called, when and by whom — because the proof that you reported immediately is the call log, not the form.

Division of responsibility: on most mines the mine itself makes the formal SAMRASS report. The contractor's product job is to capture fast, classify correctly, evidence properly, notify the mine and representatives, and run its own clocks. **Confirm with each target mine how contractor incidents are reported** before designing the hand-off.

4.7 Signature workflow 4 — Risk assessment with enforced hierarchy

Section 11 requires the employer to implement control measures **in the order in which they are listed** in the Act, as far as reasonably practicable (MHSA text). A risk assessment tool that simply lists controls misses the point.

Feature: the risk assessment form presents control levels in the statutory order, requires each higher level to be considered and recorded as applied or not reasonably practicable (with a reason) before a lower level — typically PPE — can be relied on. The record then shows, on its face, that the hierarchy was followed.

Plus periodic review dates, and linkage to the relevant mine COPs.

4.8 Signature workflow 5 — Working with the mine's system

Mandatory codes of practice. Section 9 requires employers to prepare and implement COPs complying with the Chief Inspector's guidelines; failure is a contravention and a criminal offence (MHSA COP guideline, gov.za (PDF)). Guidelines cover, among others, **trackless mobile machines** (DMPR TMM guideline (PDF)), **confined spaces**, **self-contained self-rescuers** and **flammable gas and coal dust in collieries** (GoLegal on the 2020 guidelines), and **emergency preparedness**. Compliance specialists maintain registers of current guidelines (LCS MCOP register).

The contractor does not write the mine's COPs. It works to them. So the product holds a **COP register per mine**: which of that mine's COPs apply to the contractor's scope, which version, and which workers have been trained on or acknowledged each. When a mine revises a COP, affected workers are flagged for re-acknowledgement.

Health and safety representatives. Representatives may inspect working places, identify hazards, direct employees to leave dangerous working places, and must be notified of accidents and inspections (section 25 annotated). The product links representatives to working places, notifies them automatically when the Act requires it, and lets them countersign examinations.

Section 54 instructions. Section 54 covers the inspector's power to deal with dangerous conditions (MiningSafety) — the widely known "section 54" stoppage. A **section 54 log** records instructions affecting the contractor's work, stoppage start and end, affected workers and equipment, and the evidence of remediation. That is safety evidence and — because stoppages mean lost shifts — commercial evidence too.

4.9 Signature workflow 6 — Hazardous work records and worker history

This is where the Mining edition can do something genuinely good for workers, and where it must be most careful.

Why it matters. Occupational lung disease in mining — silicosis and tuberculosis above all — is compensated under **ODMWA**, administered by the **Medical Bureau for Occupational Diseases (MBOD)**, a directorate of the National Department of Health, and the **Compensation Commissioner for Occupational Diseases (CCOD)** (UCT ODMWA module). If a mine medical practitioner finds signs of occupational lung disease at an entry, exit or periodic examination, the findings go to the MBOD, and ex-miners are entitled to free benefit medical examinations every two years (Qhubeka Trust). These diseases appear years after exposure, and claims depend on proving the work history. South Africa's silicosis compensation history shows how often that history is lost (GroundUp, OLD Collaboration).

What the product holds:

Record	Holds	Does not hold
Hazardous-work service record (s14)	Which worker did which category of hazardous work, at which mine, between which dates	Any exposure measurement results or medical findings
Exposure category	A category label from the risk assessment, e.g. high-dust work	Individual dose or sampling data
Medical surveillance status	Examination due and done dates, fit status and expiry	Anything the OMP recorded clinically
Exit record	That an exit examination was done, when, and where the certificate is held	The certificate's contents

A contractor's workers move between mines constantly. A **durable, portable record of which hazardous work each person did, where and when** — kept by the contractor, exportable to the worker — may be the only continuous work history that survives. That is worth building, and worth building carefully.

Two cautions. First, the division of medical surveillance obligations between the mine and the contractor must be confirmed with a mining SHE practitioner before this is designed; the mine's occupational medical practitioner holds the clinical record. Second, even the "does not hold" column is health-adjacent information about identifiable people. It sits under the same section 32 analysis and the same controls as the IOD documents: separate key, restricted role, never visible to the mine beyond what the worker's own compliance status requires.

4.10 Signature workflow 7 — Shift examinations and shift handover

Start-of-shift working place examinations, configured to the mine's COP requirements: examined by the supervisor at the start of the shift, photo mandatory on any unsafe condition, an unsafe finding triggering barricading, withdrawal of workers, and a corrective action. Countersignature by the health and safety representative where the mine's procedure requires it.

Shift handover log: conditions, open hazards, incomplete work and equipment status passed from outgoing to incoming supervisor, acknowledged by both.

Trigger action response plans (TARPs) — logging observed trigger levels and the actions taken — are industry practice rather than statute, and each mine defines its own. Treat them as **M3**, configured per mine.

The device constraint underground

Underground capture has a hard physical constraint that must be verified before any mining feature is designed: **many mines restrict or prohibit personal electronic devices underground, and in mines with flammable gas, electrical apparatus taken underground generally has to be certified as safe for that atmosphere.** I have not verified the specific regulations here — confirm them with a mining SHE practitioner and each target mine.

Design implications regardless of the answer:

- Offline for an **entire shift** is the normal case, not an edge case
- Support capture on **mine-approved devices** where a mine permits them
- Support **surface capture** at start and end of shift, with the shift examination recorded on surface from the underground supervisor's notes
- Support **paper-to-photo** capture: a printed examination sheet completed underground and photographed on surface, with the photograph entering the evidence chain
- Never design a workflow that assumes a phone at the working face

4.11 Compensation: RMA and the IOD engine

Rand Mutual Assurance is one of only two mutual associations licensed to provide COIDA benefits outside the Compensation Fund — the other being FEM for construction (GroundUp). Occupational injuries in mining are typically paid by RMA (PMG presentation). RMA has also administered benefits for the metal and engineering sector since 2015 (Daily Maverick / GroundUp) — relevant because many mining contractors are engineering firms.

The IOD engine runs unchanged with insurer = RMA. What must be verified directly with RMA before building: which forms RMA uses and in what format, its submission channel, whether any integration exists, and how its Letter of Good Standing is issued and renewed.

ODMWA claims are out of scope for claim management. They are medical, driven by the mine's occupational medical practitioner and the MBOD. SiteProof's contribution is the work-history record in 4.9 — not a claims workflow.

4.12 What the Mining edition must never do

Same reasoning that dropped permit-to-work from the plan: **SiteProof does not touch safety-critical control systems.**

- No gas or ventilation monitoring
- No blasting control or explosives management
- No seismic monitoring
- No collision-avoidance or proximity detection for trackless mobile machines
- No self-rescuer or lamp-room management
- Nothing that certifies, clears or releases equipment or a working place as safe

SiteProof **records** what people did and saw. It never tells anyone that a place or machine is safe. Every workflow above is a record of a human judgement, with that human named.

4.13 Mining vocabulary

Use	Not
Mine, shaft, section, working place	Site, area

Shift	Day
Mine manager	Site manager
Supervisor / shift boss	Foreman
Principal Inspector of Mines; Chief Inspector of Mines	Inspector (generic)
Mandatory code of practice (COP)	Procedure
Health and safety representative	Safety rep (informal is fine in UI copy)
Right to withdraw / section 23	Refusal
Section 54 instruction	Stop order
Occupational medical practitioner (OMP)	Company doctor
Trackless mobile machine (TMM)	Vehicle
Self-contained self-rescuer (SCSR)	Rescue pack

4.14 Validation for the Mining edition

Only after the Construction edition is paying. Talk to:

- **Three mining contractor SHE managers** — "Walk me through mobilising ten people to a new mine. How long does it take, and what goes wrong?" "How many mines do you currently serve?" "What does each mine ask for that the others don't?"
- **Two mine contractor-management coordinators** — "What makes a contractor easy to onboard? What format do you want records in?"
- **One health and safety representative** — "How are section 23 withdrawals recorded today?"
- **One occupational medical practitioner** — the division of surveillance records between mine and contractor.
- **RMA** — forms, submission channel, integration, Letter of Good Standing.

Exit signal: a contractor SHE manager says the multi-mine readiness matrix would save them days per mobilisation.

PART 5 — IMPLEMENTATION

5.1 Stack

Keep: Go (Gin, pgx) · PostgreSQL · Expo / React Native · Next.js · S3 in af-south-1 with Object Lock · Redis · JWT with refresh rotation · GitHub Actions with ECR · Docker Compose deployment · SES.

Payments: Stitch behind the `PaymentProvider` interface (1.8).

Cut: Python AI services, event bus, gRPC, microservices, OpenTelemetry, OIDC. **Biometrics permanently.**

Reusable from existing work: offline-first SQLite manifest and delta sync; JWT auth; the flag-based permission model; QR/NFC and HMAC primitives for worker cards and evidence chains; document verification flows; SAST time handling; push notifications; k6 load testing; and the lesson that ingress, not the Go backend, was the bottleneck — so media goes direct to S3 via presigned multipart.

5.2 Code structure

Mirror the three layers in the repository so the rule in 1.2 is enforced by structure, not discipline:

```
/core
  /auth /tenancy /evidence /sync /deadlines /checklists
  /actions /incidents /workers /attendance /vault /iod
  /reports /notifications /billing /audit
/regimes                      -- data: seeds, templates, taxonomies
  /za_construction_2014
  /za_mhsa_1996
/editions
  /construction
    /safetyfile /reg4 /appointments /s37 /diary /claims /programme
  /mining
    /readiness /withdrawals /shiftexam /handover /copregister
    /hazardouswork /s54log /riskhierarchy
```

Enforce it: a lint or CI check that fails the build if anything under `/core` imports from `/editions`. That single check is what keeps the platform from rotting into an `if edition ==` tangle.

5.3 Core data model

```
organisations        id, name, reg_no, vat_no, primary_edition,
                     enabled_editions[], coida_insurer
sites                id, org_id, regime_id, name, location, boundary,
                     status, principal_party_id
site_locations       id, site_id, parent_id, kind, name
                     -- construction: zone / floor
                     -- mining: shaft / level / section / working place
parties              id, site_id, role, org_name, contact, is_notice_recipient
workers              id, org_id, full_name, id_number_enc, trade, employer_party_id
worker_credentials   id, worker_id, type, number, issuer, issued_at, expires_at, file_id
worker_site_clearance id, worker_id, site_id, induction_at, induction_version,
                     induction_expires_at, medical_status, medical_expires_at,
                     short_permit_expires_at, cleared
attendance_events    id, worker_id, site_id, direction, method, device_ts,
                     server_ts, lat, lng, geofence_ok, recorded_by, corrects_id
photos               id, site_id, content_sha256, chain_prev, chain_hash,
                     device_ts, server_ts, lat, lng, mock_location, attestation
evidence_links       subject_type, subject_id, evidence_type, evidence_id
checklist_templates / items / instances / responses
corrective_actions
incidents            id, site_id, regime_code, taxonomy_code, occurred_at, ...
iod_claims / iod_claim_documents / iod_claim_ledger / iod_claim_events
deadlines            id, subject_type, subject_id, rule_code, basis_date,
                     due_at, calendar_id, satisfied_at, lapsed_at
statutory_documents  id, org_id, doc_type, issuer, reference, issued_at, expires_at
```

`worker_site_clearance` is the key generalisation. It's a thin table in Construction and the heart of the Mining edition's readiness matrix.

5.4 Mining-specific tables

```
mine_requirements     id, site_id, requirement_kind, doc_or_credential_type,
                      applies_to_role, validity_days, source_note
mine_cops             id, site_id, cop_title, version, effective_at,
                      applies_to_scope
cop_acknowledgements  id, cop_id, worker_id, acknowledged_at, method, trainer
withdrawals           (see 4.4)
shift_examinations    id, site_id, location_id, shift, examined_by,
                      examined_at, capture_mode,   -- device | surface | paper_photo
                      hs_rep_countersign_id, status
shift_handovers       id, site_id, location_id, from_supervisor, to_supervisor,
                      conditions, open_hazards, acknowledged_at
s54_instructions      id, site_id, issued_at, inspector_name, scope,
                      stoppage_start, stoppage_end, remediation_evidence_ids[]
hazardous_work_record id, worker_id, site_id, work_category,
                      started_at, ended_at, source_risk_assessment_id
exit_examinations     id, worker_id, site_id, done_at, certificate_held_by
risk_assessments      id, site_id, hazard, controls_json,  -- ordered by statutory hierarchy
                      hierarchy_justification, review_due_at
```

`hazardous_work_record` and `exit_examinations` sit under the health-data controls in 2.10.

5.5 Phase plan

Construction track

Phase	Weeks	Contents	Exit
P0 Validation	1–12	Read; interview CHSOs, subs, PCs	A CHSO would put their client base on it
P1 Core + Construction safety	12–30	Entire core; construction checklists, housekeeping, toolbox talks, hazards, incidents with W.Cl.2 clock, appointments, s37(2), MCoF, Reg 4, safety file, readiness view, Stitch billing. **Regime model and edition structure from day one.**	A CHSO runs clients on it, or a PC mandates it
P2 Construction claims	30–48	Full IOD lifecycle; JBCC delay claims; CVI; programme charts	IOD money recovered, or a notice clock saves a claim
P3 Construction multi-party	48–66	PC tier, free sub seats, rollups, GCC/NEC	A PC pays for a site licence

Mining track

Phase	When	Contents	Exit
M0 Demand signal	From P1 launch	"Mining — join the waitlist" on the signup screen	Waitlist volume and quality
M0 Validation	Once Construction is paying	Section 4.14 interviews; RMA; target mines' requirements; underground device rules	A contractor SHE manager commits to a pilot
M1 Mining core	~16 weeks	Mining regime pack, site location hierarchy, **readiness matrix**, per-mine clearance and alerts, mine requirements, pre-arrival packs, SAMRASS-aligned incidents with 10/30-day clocks and scene preservation, **right-to-withdraw register**, risk assessment hierarchy, RMA routing, surface and paper-to-photo capture	A pilot contractor mobilises to a mine using the readiness pack
M2 Mining depth	~12 weeks	COP register and acknowledgements, shift examinations and handover, H&S rep linkage, section 54 log, hazardous-work records, exit tracking	Pilot contractor uses it daily across two mines
M3 Mining	Later	TARPs per mine, mine-side read-only portal, mine-requested exports	A mine asks for access

extras			

Why Mining waits: it doubles the domain you personally have to learn, and a second edition built before the first is paying spreads a solo founder across two unproven products. The core is built edition-aware from P1, so waiting costs nothing architecturally.

5.6 Offline and devices by edition

	Construction	Mining
Typical offline duration	Minutes to hours	A full shift underground
Capture device	Worker's or foreman's phone	Mine-approved device, surface capture, or paper-to-photo
Sync point	Anywhere with signal	Usually surface
Clock-in	Gate scan by foreman	Gate or shaft-head scan; mine systems may already record access

5.7 Security specifics for Mining

Everything in 2.13, plus: hazardous-work and exit records under the health-data key; principal (mine) access limited to per-worker clearance status, never history; retention policy set with advice, given the MHSA requirement that certain records be kept until the mine closes.

PART 6 — COMPLETE FEATURE INVENTORY

Three inventories: the **shared core** (used by both editions), the **Construction edition**, and the **Mining edition**. Phase markers: **P1–P4** for core and Construction; **M1–M3** for Mining; ✕ deliberately excluded.

Where a core feature ships in P1, the Mining edition inherits it at M1 with mining regime data.

6.A Shared core

A.1 Platform

Feature	Phase	What it does
Organisation accounts	P1	Registration, VAT, COIDA insurer, primary and enabled editions
Edition selection at signup	P1	Routes to the edition's dashboard and onboarding
Edition attached to site	P1	Each site uses one regime; organisations can hold both editions
Regulatory regime model	P1	All regime-specific content held as data
Core/edition import boundary check	P1	CI fails if core imports edition code
User accounts, JWT with refresh rotation	P1	Standard authentication
Device session revocation	P1	Lost devices cut off
Phone-number invitations	P1	SMS deep-link invites
Roles and capability flags	P1	Defaults per role; individual capabilities grantable
Site scoping	P1	Users see only assigned sites
Site location hierarchy	P1	Zones and floors, or shafts, levels, sections and working places
Onboarding wizard per edition	P1	Edition-specific first-run setup
Mining waitlist on signup	P1	Demand signal for the second edition
Multi-organisation membership	P2	Consultants and CHSOs across client companies

A.2 Evidence

Feature	Phase	What it does
In-app capture with GPS	P1	Location, device time, site, user, device
On-device compression	P1	2048px long edge before hashing
SHA-256 at capture	P1	Re-verified server-side
Dual timestamps	P1	Divergence shown honestly
Per-site hash chain	P1	Insertion or deletion detectable
Immutable originals, derived markup	P1	Originals never altered
Watermarking	P1	Company, site, reference, date, GPS
Tagging and galleries	P1	Filterable evidence library
Access log	P1	Every view, export and share
Paper-to-photo capture	P1	Completed paper forms enter the evidence chain
Daily signed chain anchor	P2	Separate append-only store
Device attestation	P2	Play Integrity, App Attest
Mock-location flagging	P2	Recorded, not discarded
Video, voice notes, document scanning	P2	Same integrity treatment
Evidence export pack	P2	Hash manifest and chain verification

A.3 Offline and sync

Feature	Phase	What it does
Full offline capture	P1	All field functions without signal
Full-shift offline tolerance	P1	Eight to twelve hours offline treated as normal
Client-generated UUIDs	P1	Idempotent sync
Separate metadata and media queues	P1	Records before photos
Resumable multipart uploads	P1	Resume, not restart
WiFi-preferred with data override	P1	Size estimate before mobile data use
Local storage budget and warnings	P1	No silent loss
Per-record sync status	P1	Users can see their evidence is safe

A.4 Deadline engine

Feature	Phase	What it does
Calendar and working-day rules	P1	Per-rule day basis
SA public holiday calendar as data	P1	Including the Sunday-to-Monday rule and moving dates
Site shutdown periods	P1	e.g. the builders' holiday
Immediate-report prompts	P1	Call-now prompts with call logging, not countdowns
Recurring cadences	P2	e.g. monthly medical progress reports
Calendar inspector	P1	Shows excluded days and why
Computation and notification log	P1	Audit trail
Escalation up the role hierarchy	P1	To the owner on lapse
Non-completion alerting	P1	Detects a silently stopped job

A.5 Safety engines

Feature	Phase	What it does
Versioned checklist engine	P1	Templates from regulatory packs
Photo mandatory on fail	P1	No non-compliance without evidence
Reference photos	P1	What "correct" looks like
Per-shift, daily and per-activity scheduling	P1	Both editions' cadences
Custom checklist builder	P2	Organisation-specific additions
Corrective action register	P1	Owner, due date, before/after photos, verification
Open-items ageing	P1	The core safety metric
Hazard identification	P1	Location, risk before/after, mitigation
Near-miss one-tap capture	P1	Thirty-second, no-blame
Toolbox talks with content library	P1	Per-edition content
PPE checks and issue register	P1	Proof of issue
Risk assessments with review dates	P1	Versioned
Induction records	P1	Content version and expiry

A.6 Incidents

Feature	Phase	What it does
Incident capture	P1	Type, people, injury, witnesses, photos, sketch
Regime taxonomy	P1	OHSA or SAMRASS-aligned classification
Obligation routing	P1	Every reporting path the regime triggers
Incident register	P1	Maintained automatically
Investigation workflow	P2	Root cause, actions, close-out

A.7 Workers and attendance

Feature	Phase	What it does
Worker register	P1	Encrypted ID, trade, employer, photo
Credentials and competencies	P1	With expiry
Medical fitness status and expiry	P1	Never contents
Site-specific clearance	P1	Worker cleared per site
QR/NFC worker cards	P1	Supervisor scan, offline
Roll-call fallback	P1	Always available
Geofence check and headcount reconciliation	P1	Buddy-punching controls
Append-only attendance	P1	Corrections as new rows
Clock-in block on lapsed requirements	P1	Medical, induction, credential
Attendance reporting and payroll export	P2	Export only
Biometric attendance	×	Prohibited category under POPIA s26

A.8 Statutory vault

Feature	Phase	What it does
Organisation document vault	P1	LGS, tax, insurances, B-BBEE, registrations
Issuer-aware Letter of Good Standing	P1	FEM, RMA or Compensation Fund
Expiry alerting at 60/30/7 days	P1	Escalating
Annual filing reminders	P1	Return of Earnings, renewals
Verification status	P2	Uploaded → verified → expired → rejected

A.9 Injury-on-duty claims

Feature	Phase	What it does
Pre-filled employer's report	P1	From incident, worker and organisation data
Doctor handout	P1	Shareable by WhatsApp or print
7-day reporting clock	P1	Escalating
Insurer routing	P1	FEM, RMA or Compensation Fund
Claim stage tracker	P2	Reported → accepted → treatment → resumed → closed
Document checklist with ageing	P2	Every required form and supporting document
Account-free doctor upload link	P2	Expiring, upload-only
Monthly progress-report cadence	P2	While the worker is off
Claim number capture	P2	From the acknowledgement card

Attendance-triggered resumption report	P2	Worker clocks in → report prompted
Light duties tracking	P2	Offered and performed
Reimbursement ledger	P2	Money owed on open claims
Submission pack export	P2	Ordered for portal upload
Claims-handler role	P2	Only role with health documents
Filing on the employer's behalf	×	The employer remains the reporting party

A.10 Reports, notifications, billing, security

Feature	Phase	What it does
Report engine with versioned templates	P1	Reproducible years later
Customer branding	P1	Their letterhead
Asynchronous large-pack generation	P1	Safety files and readiness packs
Push, email, in-app notifications	P1	Deadlines, expiries, incidents, sync failures
SMS for critical alerts	P2	Paid add-on
Stitch DebiCheck with card fallback	P1	Default billing
Provider-agnostic billing layer	P1	Plans, mandates, invoices, attempts
VAT invoices, annual prepay, dunning	P1	Standard billing
Mandate revocation on cancel	P1	No further collection possible
Free read-only tier on lapse	P2	Evidence stays readable
Tenant isolation plus row-level security	P1	Two lines of defence
Separate keys: evidence, app, health	P1	Independent key control
S3 Object Lock	P1	Evidence cannot be deleted
Append-only audit log	P1	Shipped off-instance
Tested quarterly restore	P1	Backups proven
Data export on request	P2	POPIA rights

6.B Construction edition

Feature	Phase	What it does
Construction Regulations checklist pack	P1	Pre-mapped templates
Construction toolbox talk content	P1	Activity and season based
Housekeeping walks by zone	P1	Scored with photos
Housekeeping trends	P1	Per-site trend lines
Appointments register and letters	P1	Construction manager, supervisor, competent persons
Appointment gap detection	P1	Activity with no appointment
Section 37(2) mandatary agreements	P1	Register, generation, signatures
Regulation 4 notification	P1	7-day deadline and tracking
H&S plan and specification storage	P1	Versioned
MCoF Annexure 3 tracking	P1	Status and one-year expiry
Annexure 3 employer-section pre-fill	P2	From trade and risk assessment
OHSA s24 reportable incident flow	P1	Annexure 1 preparation
W.Cl form set for IOD	P1–P2	W.Cl.2 to W.Cl.6 templates

FEM routing and LGS	P1	Default construction insurer
Safety file assembly	P1	Inspector order
Inspection readiness view	P1	Present, missing, expired
Site diary, two-tier	P1	Sixty-second default, escalation
Weather by effect	P1	Activities stopped, hours lost, delayed effect
Labour by trade and subcontractor	P1	Hours worked and standing
Plant working vs standing	P1	Recoverable standing time
Materials expected not received	P1	Delay ground
Construction work permit tracking	P3	Above-threshold projects
Activity-specific plans	P3	Fall protection, excavation, lifting
JBCC contract form, edition-aware	P2	Drives notice rules
Contract data capture	P2	Penalties, builders' holiday, notice method
Delay claims register	P2	Clocks and value at risk
Three-clock state machine	P2	Notice, submission, PA response, deemed rejection
Notices from customer's own domain	P2	With proof of service
Confirmation of verbal instruction	P2	Formal CVI letter
Evidence linking and quantum entry	P2	Many-to-many, itemised
Submission and dispute packs	P2	For QS or attorney
Programme chart rendering	P2	As-planned vs as-built, server-side SVG
Principal contractor site licence	P3	With free sub seats
PC compliance rollup	P3	Across all subs
Incoming claim tracker for PCs	P3	With response clocks
Visibility boundary screen	P3	Subs see exactly what the PC sees
GCC and NEC contract forms	P3	Additional notice regimes
Variations register	P3	Against contract value
Critical-path calculation	P4	Scheduling engine
CIDB grading evidence pack	P4	Verify requirements first

6.C Mining edition

Feature	Phase	What it does
MHSA regime pack	M1	Checklists, taxonomy, deadlines, vocabulary
Mine location hierarchy	M1	Shaft, level, section, working place; surface vs underground
Readiness matrix	M1	Every worker × every mine served
Per-mine requirements	M1	Each mine's inductions, medicals, competencies, documents
Per-mine clearance and expiry alerts	M1	A worker can lapse at one mine only
Short-duration permit tracking	M1	Where a mine allows brief work before full clearance
Pre-arrival readiness pack	M1	Exported to the mine's contractor office
RMA routing and Letter of Good Standing	M1	Default mining insurer
SAMRASS-aligned incident taxonomy	M1	Classification from the published codebook
Dangerous-occurrence call-now prompt	M1	Immediate reporting with call log

10-day investigation start clock	M1	MHSA s11(5)(aA)
30-day investigation report clock	M1	To Principal Inspector and H&S committee
Heat-injury SAMRASS 8 clock	M1	14 days
Scene-preservation flag	M1	Fatal, likely fatal or loss of limb
Right-to-withdraw register	M1	Section 23 withdrawals, notification, resolution
Risk assessment with enforced hierarchy	M1	Controls in the statutory order
Surface and paper-to-photo capture	M1	For shifts without devices underground
Mine-approved device support	M1	Where a mine permits capture underground
H&S representative linkage	M2	Reps per working place; automatic notifications
H&S rep countersignature	M2	On examinations where the mine requires it
COP register per mine	M2	Applicable COPs, versions, scope
COP acknowledgements	M2	Per worker; re-flag on revision
Start-of-shift working place examinations	M2	Per the mine's COP
Unsafe-finding flow	M2	Barricade, withdraw, corrective action
Shift handover log	M2	Outgoing to incoming supervisor
Section 54 instruction log	M2	Instructions, stoppages, remediation evidence
Hazardous-work service record	M2	Work category, mine, dates — no measurements
Exit examination tracking	M2	That it happened, not its contents
Worker work-history export	M2	Portable record for the worker
Mining toolbox talk content	M2	Mining-specific topics
TARP logging per mine	M3	Trigger levels and actions
Mine-side read-only portal	M3	Clearance status for the mine
Mine-specific export formats	M3	Whatever each mine's system needs
ODMWA claim management	×	Medical, OMP- and MBOD-driven
Gas, ventilation, seismic, blasting systems	×	Safety-critical control systems
TMM proximity or collision avoidance	×	Safety-critical control systems
Clearing or certifying a place as safe	×	SiteProof records judgements; never makes them

6.D Totals

Inventory	P1 / M1	P2 / M2	P3 / M3	P4	Excluded	Total
Shared core	85	24	—	—	2	111
Construction	20	10	8	2	—	40
Mining	18	12	3	—	4	37

Construction's P1 count includes the W.Cl form set, which spans P1 and P2.

The execution risk is P1. It contains the whole core plus the construction safety layer. If it must shrink, keep: checklists with photo-on-fail, corrective actions, incidents with the reporting clock, the worker register with attendance and clearance, the statutory vault, offline capture, and the safety file. Defer housekeeping trends, appointment gap detection and s37(2) to early P2.

PART 7 — RESOURCE LIBRARY

7.1 Read these first

Construction

1. Construction Regulations 2014 (PDF)
2. Labour Guide — Guide in respect of COIDA claims
3. JBCC Advisory Note on Extension of Time (PDF)

Mining

1. Mine Health and Safety Act and Regulations — MHSC booklet (PDF)
2. Regulation 23 — Accidents and dangerous occurrences
3. DMR guideline — Minimum standards of fitness to perform work at a mine (PDF)

Both

- POPIA section 32 and the 2026 Health Information Regulations explained

7.2 Shared: data protection and evidence

- Information Regulator of South Africa
- POPIA full text · Section 26 · Section 32
- ITLawCo — Health Information Regulations 2026
- Bowmans — Health Information Regulations cross the finish line
- Labour Guide — what employers must know
- Mayet & Associates — new POPIA regulations on health information
- Moonstone — the scope debate
- ASISA POPIA guidelines (PDF)
- For **ECTA**, read section 15 of the Electronic Communications and Transactions Act 25 of 2002 at source.

7.3 Shared: COIDA and injury claims

- Labour Guide — Claiming procedure for injuries on duty
- Labour Guide — Guide in respect of claims
- Labour Guide — COID form downloads
- Compensation Fund Assistance — full form list
- Bowmans — correctly submitted injury claims
- Evergreen Accounting — worked COIDA claim example
- EOHCB — employer reporting obligations
- EOHCB — occupational disease sequence
- Pay Solutions — claiming procedures
- CompEasy · Return of Earnings

7.4 Construction

Health and safety law

- Construction Regulations 2014 (PDF) · ASAQS guidelines (PDF) · Master Builders guidelines (PDF)
- Regulation 4 — Notification · Polity — Construction Regulations 2014 · Labour Guide — permit exemption
- CIFA — impact on clients and designers (PDF)
- MAKROSAFE — OHSA section 24 reporting · Gawie le Roux Institute — injury on duty
- Department of Employment and Labour · SAIOSH · Labour Guide

Medical fitness

- Annexure 3 (PDF) · Annexure 3 example (PDF) · Occupational Health SA — MCoF (PDF) · CIDB primary health standard (PDF)

FEM and the Letter of Good Standing

- FEM · FEM — construction benefits · South African Builder · Tenderkom · ClearComply

Contracts and delay claims

- JBCC · EOT advisory note (PDF) · Advisory note 23.0 (PDF)
- Cox Yeats · Hahn & Hahn · Le Roux Consulting · Agora Africa
- Lexology — JBCC subcontract EOT · Lexology — GCC weather · SciELO — JBCC claim events
- SAICE · NEC · Society of Construction Law
- **Book:** *Construction Delay Analysis Simplified — The South African Edition*, Dr Hendrik Prinsloo

Industry bodies

- CIDB · Register of Contractors · SACPCMP · ASAQS · Master Builders SA · SASOHN

7.5 Mining

The Act and Regulations

- Mine Health and Safety Act — gov.za
- MHSA consolidated — SAFLII
- MHSA text (CER, PDF) — sections 9 to 17, 23 and 25 in full
- Mine Health and Safety Regulations — SAFLII — all 23 chapters
- MHSC — Act and Regulations booklet (PDF)
- Section 23 — right to leave dangerous working place
- Section 25 — health and safety representatives
- Barter McKellar — guide to mine health and safety law

Reporting

- Regulation 23 — accidents and dangerous occurrences
- SAMRASS codebook (PDF) — the classification to load into the incident taxonomy
- MHSC — incident reporting protocol example (PDF)

Codes of practice

- MHSA COP guideline example — gov.za (PDF) — shows the legal status of COPs
- DMPR — TMM COP guideline (PDF)
- SCSR COP guideline, GNR 27 of 2020 (PDF)
- GoLegal — the 2020 COP guidelines
- LCS — MCOP register · MCOP register (PDF)
- DMR — Emergency preparedness and response guideline (PDF)

Medical fitness and contractors on mines

- DMR — Minimum standards of fitness to perform work at a mine (PDF)
- MiningSafety — contractors, medicals and inductions on mines

Compensation: RMA and ODMWA

- GroundUp — the shifting sands of compensation for sick miners — RMA, FEM and ODMWA explained
- GroundUp — the driller from Lesotho — why work history matters
- UCT — ODMWA, MBOD and CCOD
- Qhubeka Trust — mineworkers' rights under ODMWA
- OLD Collaboration — silicosis compensation
- Southern Africa Trust — portability of mining social security benefits (PDF)

Competitors and adjacent tools

- OPMS — mining onboarding · Kynection · Induct For Work · Quartex — mine-side onboarding, mostly Australian
- Ariscu — SA contractor safety management
- SmartQHSE — mining HSE, claims MHSA registers

- OVS Solutions — SA safety software landscape

Bodies to know — search for each directly: the Mine Health and Safety Inspectorate (Department of Mineral and Petroleum Resources), the Mine Health and Safety Council, the Mining Qualifications Authority, the Minerals Council South Africa, Rand Mutual Assurance, and the Medical Bureau for Occupational Diseases.

7.6 Payments

- Stitch — recurring payments · Debit orders and DebiCheck guide
- Stitch docs — DebiCheck introduction · DebiCheck integration
- Peach Payments — recurring billing — the alternative, if ever needed

7.7 Technical

- Go · pgx · PostgreSQL · Row-level security
- Expo · expo-sqlite · expo-file-system · expo-location
- S3 multipart · Presigned URLs · Object Lock · AWS pricing calculator
- Play Integrity · App Attest · SES event publishing · OWASP ASVS

PART 8 — RISKS AND CAVEATS

8.1 The biggest risk is attention

Two editions of a regulated safety product is a large commitment for a solo founder already running a live platform. The phase plan deliberately makes Mining wait until Construction is paying. Hold that line.

8.2 Risk register

Risk	Edition	Severity	Mitigation
Deadline engine gives wrong advice	Both	**Severe**	Professional review; show workings; never authoritative; log everything
Mishandling health information	Both	**Severe**	s32(1)(f) basis confirmed; separate key; claims-handler role; never visible to principals
Health-adjacent mining records exposed to mines	Mining	**Severe**	Mines see clearance status only, never history
Being mistaken for a safety control system	Mining	**Severe**	Never clear, certify or release anything; section 4.12 exclusions
Underground device rules make field capture impossible	Mining	**High**	Verify first; surface and paper-to-photo capture from M1
Building Mining before Construction pays	Both	High	Waitlist first; M0 validation gated on revenue
Contractor-vs-mine surveillance obligations misunderstood	Mining	High	OMP and mining SHE review before M2
Mines mandate their own systems	Mining	Medium	Export to whatever each mine requires; contractor-side value remains
Positioning as a generic safety app	Both	High	Compete on statutory specificity
Core rots into edition conditionals	Both	Medium	CI import boundary check
Regulatory thresholds, forms and editions change	Both	Medium	Everything regime-specific held as data
DebiCheck integration effort	Both	Medium	Card fallback from day one

8.3 What this document is not

Not legal advice, not a specification, and not a substitute for practitioners. Before any part becomes code, have it reviewed by:

- **Construction:** a construction attorney, a quantity surveyor, a SACPCMP-registered CHSO, and FEM
- **Mining:** a mining SHE practitioner, a mine's contractor management function, an occupational medical practitioner, and RMA
- **Both:** a privacy practitioner, on the health information position in particular

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Revision 4, compiled 22 September 2026. Scope: Construction and Mining editions only. Supersedes Revisions 1–3. Sources verified at time of writing and linked throughout. Verify at source before relying on any threshold, deadline, form, section or clause reference.